

Stochastic Interest Rate Modelling and Prediction

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1 Introduction and Motivation

Interest rates are the fundamental primitive of modern finance. They determine the present value of every future cash flow, price every fixed-income instrument, and underpin the risk management frameworks of banks, insurers, and pension funds. Yet despite their centrality, interest rates resist simple deterministic description. They exhibit mean reversion toward long-run equilibrium levels, time-varying volatility, and a complex term structure in which rates at different maturities move together but not identically. Capturing this behaviour requires mathematical models rooted in stochastic calculus.

The Cox–Ingersoll–Ross model, introduced in their landmark 1985 paper, remains after four decades one of the most important and pedagogically instructive short-rate models ever devised. Its elegance lies in a deceptively simple stochastic differential equation whose mathematical structure generates tractable closed-form solutions for bond prices, yields, and moments, while simultaneously satisfying the economically essential requirement that interest rates remain non-negative. The model is a single-factor model, meaning the entire yield curve is driven by one latent state variable, the instantaneous short rate. This is both its greatest strength and its most significant limitation, and understanding precisely where and why it succeeds or fails forms a central theme of this report.

This project approaches the CIR model from three distinct but mathematically unified perspectives. The stochastic calculus perspective derives the model from first principles, characterises the exact probability distribution of the short rate at any future horizon via the Fokker–Planck equation, calibrates parameters using full Bayesian inference with Metropolis–Hastings MCMC, and simulates future paths. The financial econometrics perspective demonstrates that discretising the CIR SDE recovers a familiar AR(1)–GARCH(1,1) specification, fits this model using Maximum Likelihood Estimation, and applies cointegration analysis and Nelson–Siegel decomposition to characterise the

level, slope, and curvature of the yield surface. The statistical learning perspective asks whether non-parametric models augmented with CIR-derived features can improve out-of-sample forecasts. A final robustness layer compares OLS, GMM, and MLE/MCMC estimators and evaluates three structural extensions of the base model.

The data consist of daily zero-coupon yields from 2016 to mid-2024, covering nine maturities. The training period spans 2016 to April 2024, and the test period covers approximately 496 trading days thereafter. The underlying market is withheld to avoid macroeconomic prior bias.

2 Data Engineering and Exploratory Analysis

2.1 Dataset Structure and Preprocessing

The raw dataset contains daily observations across nine maturity tenors: 3M, 6M, 9M, 1Y, 2Y, 5Y, 10Y, 20Y, and 30Y. Let $\tilde{y}(t, \tau)$ denote the raw observed yield at date t for maturity τ . The relationship between the observed and true yield is:

$$\tilde{y}(t, \tau) = y(t, \tau) + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \sigma_\varepsilon^2)$$

Preprocessing follows a two-pass pipeline. In the first pass, zero values are treated as missing and replaced using linear interpolation, with forward-filling applied to structural gaps. In the second pass, outliers are detected using the modified Z-score:

$$M_i = \frac{0.6745 (x_i - \tilde{x})}{\text{MAD}}, \quad \text{MAD} = \text{median}(|x_i - \tilde{x}|)$$

where $\tilde{x}$ is the sample median. Observations with $|M_i| > 3.5$ are flagged and replaced. This approach is preferred over the standard Z-score because it is robust to the very outliers it is designed to detect.

2.2 Historical Yield Time Series and Surface

Figure 1 displays the full training set. Three distinct regimes are immediately apparent: a gradual normalisation phase from 2016 to 2019, the sharp 2020 collapse to near-zero rates, and the aggressive post-2022 tightening cycle that drove short rates to approximately five percent and inverted the yield curve. Figure 2 renders the full yield curve data as a three-dimensional surface over time and maturity, capturing the time-varying level, slope, and curvature of the term structure simultaneously.



Figure 1: Historical zero-coupon yields for all nine maturities over the training sample (2016–2024). Three regimes are visible: gradual normalisation, the 2020 collapse, and the sharp post-2022 tightening cycle.

Yield Surface -- Training Data

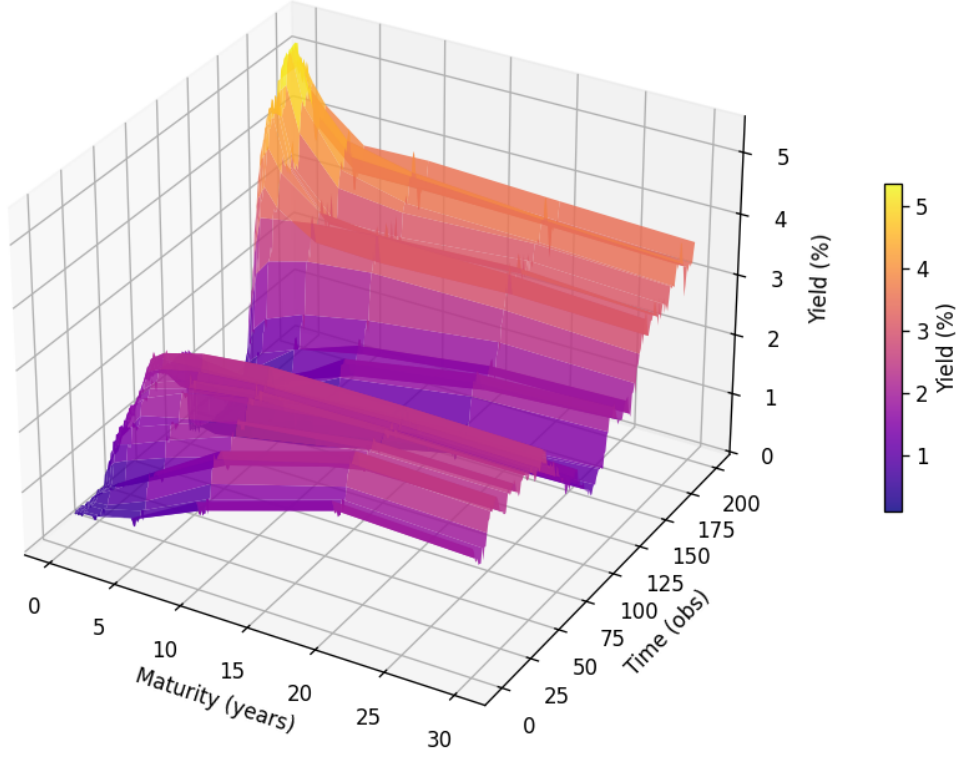


Figure 2: Three-dimensional yield surface over the training period. The surface encodes the level, slope, and curvature dimensions of the yield curve simultaneously.

2.3 Stationarity Analysis

Table 1 reports Augmented Dickey–Fuller test results for each tenor. The ADF test evaluates the null hypothesis of a unit root against the alternative of stationarity, with the test statistic being the t -ratio on the coefficient δ in the auxiliary regression $\Delta y_t = \mu + \delta y_{t-1} + \sum_{k=1}^p \phi_k \Delta y_{t-k} + \varepsilon_t$.

Table 1: Augmented Dickey–Fuller stationarity tests. All yield series are integrated of order one, I(1): non-stationary in levels but stationary in first differences.

Tenor	ADF (level)	p -value	ADF (diff)	p -value	Order
3M	0.0045	0.9589	−5.5146	0.0000	I(1)
6M	0.2715	0.9760	−9.7005	0.0000	I(1)
9M	−0.0890	0.9505	−6.2695	0.0000	I(1)
1Y	−0.2130	0.9370	−6.9366	0.0000	I(1)
2Y	0.0226	0.9604	−42.8389	0.0000	I(1)
5Y	−0.5498	0.8819	−42.9167	0.0000	I(1)
10Y	−0.6478	0.8598	−44.7937	0.0000	I(1)
20Y	−0.7762	0.8260	−22.6505	0.0000	I(1)
30Y	−0.8425	0.8063	−46.3578	0.0000	I(1)

Every tenor fails to reject the unit root hypothesis in levels ($p > 0.9$ throughout) but strongly rejects it in first differences ($p < 0.0001$), confirming all yield series are I(1). This is a critical finding for econometric modelling: levels regressions among these series are valid only if the series are cointegrated, a hypothesis examined formally in Section 5.5.

2.4 ACF and PACF of First-Differenced Yields

Figure 3 displays the autocorrelation function (ACF) and partial autocorrelation function (PACF) of first-differenced yields at four representative maturities. The 3M and 1Y series retain statistically significant autocorrelations at the first few lags, consistent with an AR(1) or AR(2) generating process. The longer maturities (5Y and 30Y) show nearly white-noise first differences. This maturity-dependent serial dependence structure has direct implications for model specification and is reflected in the GARCH estimation results in Section 5.

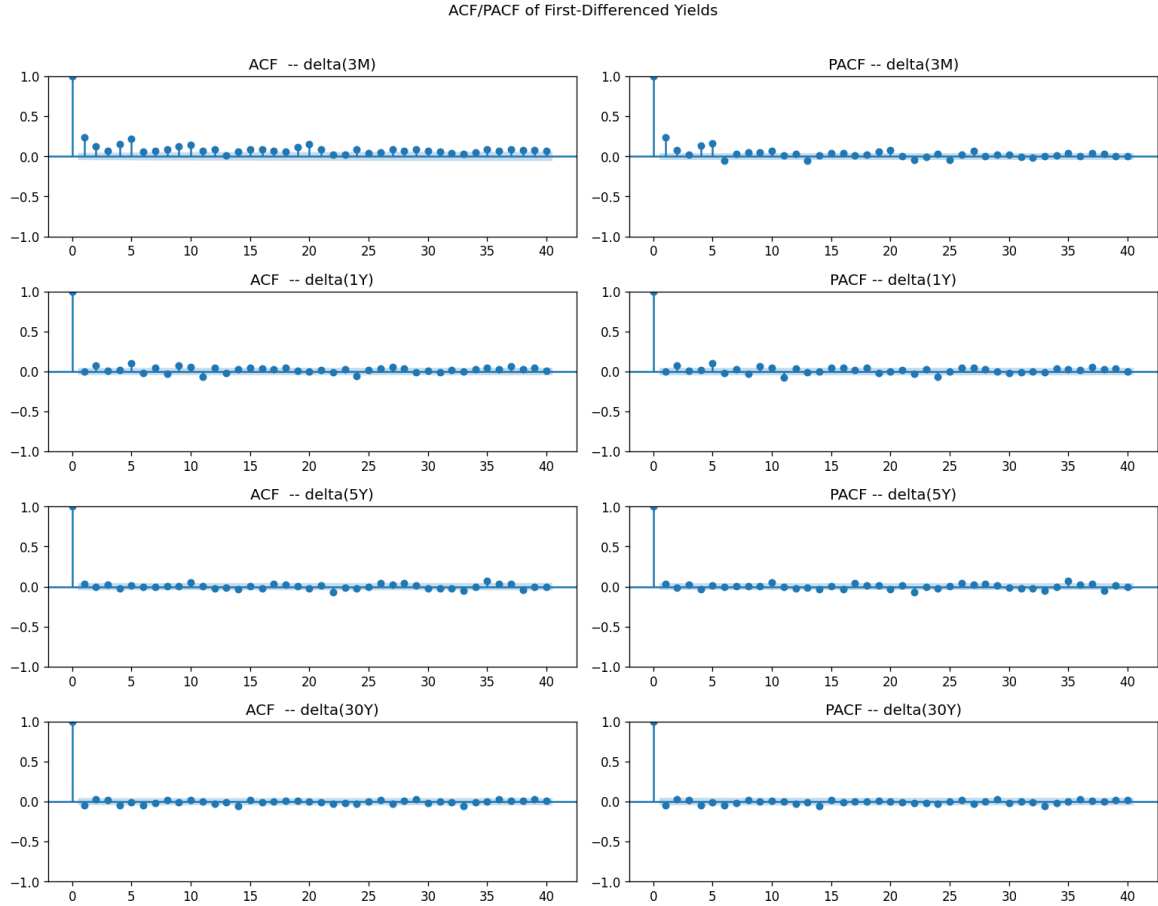


Figure 3: ACF and PACF of first-differenced yields at four maturities. The short end exhibits mild serial dependence; the long end is effectively white noise in differences.

3 Financial Mathematics: The CIR Model

3.1 Economic Derivation of the SDE

Cox, Ingersoll and Ross derive the short-rate process from a general equilibrium production economy. The representative agent maximises:

$$\mathbb{E}_0 \left[\int_0^\infty e^{-\rho t} \log(C_t) dt \right]$$

Under log utility and a production variance proportional to the current output level (a square-root specification), the equilibrium real interest rate satisfies:

$$dr_t = \kappa(\theta - r_t) dt + \sigma\sqrt{r_t} dW_t, \quad r_0 > 0$$

where $\{W_t\}$ is a standard Brownian motion. The parameter $\kappa > 0$ governs the speed of mean reversion, $\theta > 0$ is the long-run equilibrium rate, and $\sigma > 0$ controls the magnitude of random fluctuations. The diffusion coefficient $\sigma\sqrt{r_t}$ prevents the process

from becoming negative under the Feller condition.

Theorem 3.1 (Feller 1951). *The CIR process satisfies $r_t > 0$ almost surely for all $t > 0$ if and only if $2\kappa\theta \geq \sigma^2$. When this condition holds, zero is an inaccessible boundary.*

3.2 Closed-Form Moments via Ito's Lemma

Applying Ito's lemma to $f(r_t) = r_t$ and taking expectations yields:

$$\mathbb{E}[r_t | r_0] = \theta + (r_0 - \theta)e^{-\kappa t}$$

The mean-reversion half-life is $t_{1/2} = \log 2 / \kappa$. With the calibrated value $\hat{\kappa} \approx 0.047$, the half-life is approximately 14.6 years or 3,683 trading days, consistent with the very long swings visible in the historical data. The conditional variance is:

$$\text{Var}(r_t | r_0) = \frac{\sigma^2 \theta}{2\kappa} (1 - e^{-\kappa t})^2 + \frac{\sigma^2 r_0}{\kappa} e^{-\kappa t} (1 - e^{-\kappa t})$$

As $t \rightarrow \infty$, the stationary distribution is $\text{Gamma}(2\kappa\theta/\sigma^2, \sigma^2/(2\kappa))$.

3.3 Exact Transition Density

The exact transition density of the CIR process is:

$$p(r_t | r_s) = c e^{-u-v} \left(\frac{v}{u}\right)^{q/2} I_q(2\sqrt{uv})$$

where $c = 2\kappa/[\sigma^2(1 - e^{-\kappa\Delta t})]$, $u = cr_s e^{-\kappa\Delta t}$, $v = cr_t$, $q = 2\kappa\theta/\sigma^2 - 1$, and I_q is the modified Bessel function of the first kind. Equivalently, $2cr_t | r_s \sim \chi^2(2q + 2, 2u)$. The availability of this exact density is crucial: it enables an exact likelihood for parameter estimation, avoiding Euler-discretisation bias.

3.4 Fokker–Planck Equation and Density Evolution

The Fokker–Planck (Kolmogorov forward) equation governs how the probability density $p(r, t)$ evolves:

$$\frac{\partial p}{\partial t} = -\frac{\partial}{\partial r} [\kappa(\theta - r)p] + \frac{1}{2} \frac{\partial^2}{\partial r^2} [\sigma^2 r p]$$

This PDE is solved numerically using the Crank–Nicolson finite-difference scheme, which at each time step produces a tridiagonal linear system solved in $O(N)$ operations via the Thomas algorithm. Figure 4 displays the density evolution from the initial short rate toward the stationary Gamma distribution.

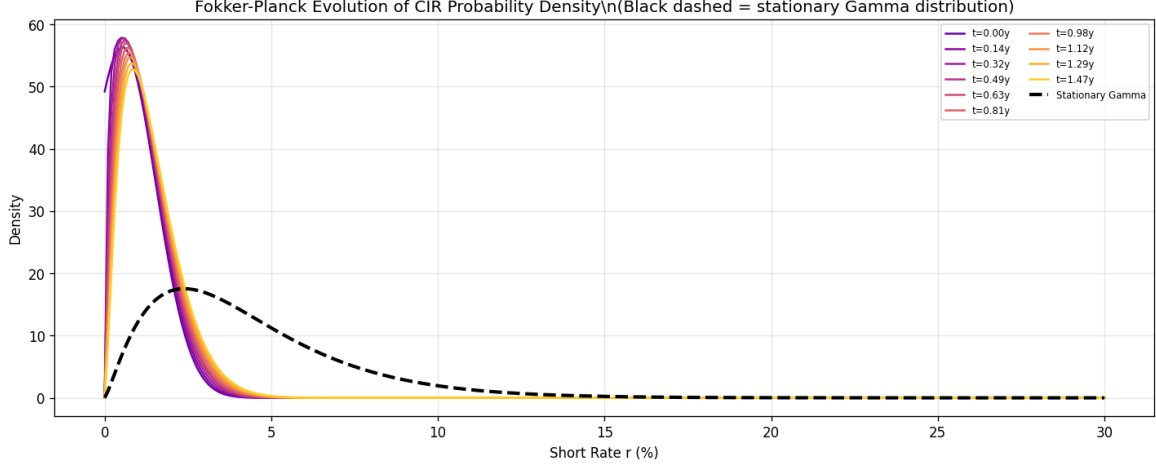


Figure 4: Fokker–Planck evolution of the CIR probability density (Crank–Nicolson). The density spreads from the initial spike and converges toward the stationary Gamma distribution (black dashed).

3.5 Edgeworth Series Approximation

The non-central chi-squared transition density is accurately approximated by a fourth-order Edgeworth expansion around a Gaussian baseline. Writing $z = (x - \mu_X)/\sigma_X$ where $x = 2cr_t$:

$$p_X(x) \approx \frac{\phi(z)}{\sigma_X} \left[1 + \frac{\kappa_3}{6\sigma_X^3} H_3(z) + \frac{\kappa_4}{24\sigma_X^4} H_4(z) + \frac{\kappa_3^2}{72\sigma_X^6} H_6(z) \right]$$

where $H_3(z) = z^3 - 3z$, $H_4(z) = z^4 - 6z^2 + 3$, $H_6(z) = z^6 - 15z^4 + 45z^2 - 15$, and cumulants $\kappa_3 = 8(q + 1 + 3u)$, $\kappa_4 = 48(q + 1 + 4u)$. Figure 5 confirms this approximation is visually indistinguishable from the exact density at a six-month horizon, validating the Gaussian-family approach.

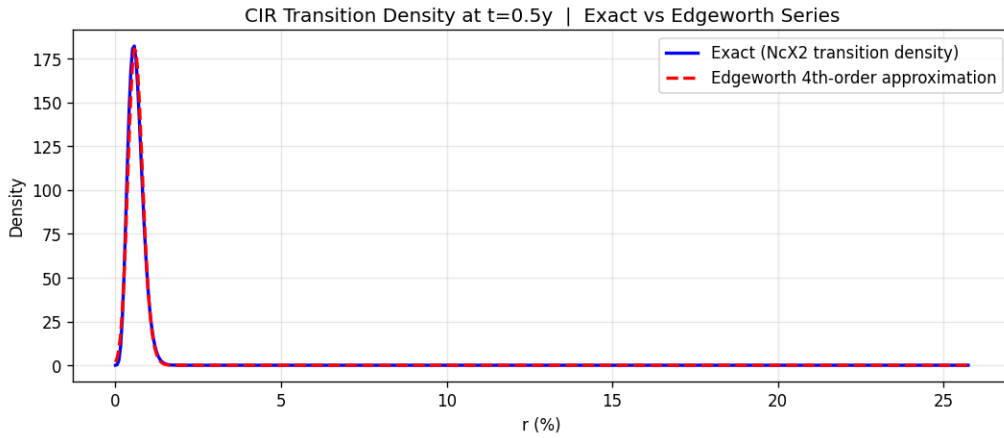


Figure 5: Exact non-central chi-squared transition density versus fourth-order Edgeworth series at a six-month horizon. The two are nearly identical.

3.6 Bond Pricing via the Feynman–Kac Formula

The zero-coupon bond price $P(t, T) = \mathbb{E}^{\mathbb{Q}}[e^{-\int_t^T r_s ds} \mid r_t]$ satisfies:

$$\frac{\partial P}{\partial t} + \kappa^*(\theta^* - r) \frac{\partial P}{\partial r} + \frac{\sigma^2 r}{2} \frac{\partial^2 P}{\partial r^2} - rP = 0, \quad P(T, T) = 1$$

with $\kappa^* = \kappa + \lambda\sigma$ and $\theta^* = \kappa\theta/\kappa^*$ under the risk-neutral measure. The closed-form solution is:

$$P(t, T) = A(\tau)^{2\kappa\theta/\sigma^2} e^{-B(\tau)r_t}$$

$$B(\tau) = \frac{2(e^{\gamma\tau} - 1)}{(\gamma + \kappa^*)(e^{\gamma\tau} - 1) + 2\gamma}, \quad A(\tau) = \frac{2\gamma e^{(\gamma + \kappa^*)\tau/2}}{(\gamma + \kappa^*)(e^{\gamma\tau} - 1) + 2\gamma}$$

with $\gamma = \sqrt{(\kappa^*)^2 + 2\sigma^2}$ and $\tau = T - t$. The continuously-compounded yield is:

$$y(\tau; r_t) = \frac{B(\tau)r_t - \log A(\tau)^{2\kappa\theta/\sigma^2}}{\tau} = \underbrace{\frac{f(\tau)}{\tau}}_{\text{base level}} + \underbrace{\frac{g(\tau)}{\tau}}_{\text{sensitivity}} \cdot r_t$$

where $g(\tau) = B(\tau)/\tau$ and $f(\tau) = -\log A(\tau)^{2\kappa\theta/\sigma^2}/\tau$. This affine structure in r_t is the key equation driving all yield curve predictions in this project.

4 Bayesian Calibration via MCMC

4.1 Estimation Strategy

Given the exact non-central chi-squared transition density, the log-likelihood for a sequence of short-rate observations $\{r_1, \dots, r_T\}$ is:

$$\log \mathcal{L} = \sum_{t=1}^{T-1} \log p(r_{t+1} \mid r_t; \kappa, \theta, \sigma, \Delta t)$$

A Bayesian approach is adopted with weakly informative priors:

$$\kappa \sim \text{Gamma}(2, 1), \quad \theta \sim \text{Gamma}(2, 0.05), \quad \sigma \sim \text{HalfNormal}(0.3)$$

The Metropolis–Hastings algorithm with log-space random-walk proposals runs for 10,000 iterations with 3,000 burn-in, achieving an acceptance rate of approximately 31%.

4.2 MCMC Diagnostics and Estimates

Figure 6 displays trace plots, posterior marginals, and chain ACF for all three parameters. The posterior for σ is sharply concentrated (well-identified from daily return

data), while κ and θ show slower mixing due to their high posterior correlation. The posterior median estimates are:

$$\hat{\kappa} = 0.0474, \quad \hat{\theta} = 4.29\%, \quad \hat{\sigma} = 0.0425$$

The Feller condition is satisfied: $2\hat{\kappa}\hat{\theta} = 0.00407 > \hat{\sigma}^2 = 0.00181$.

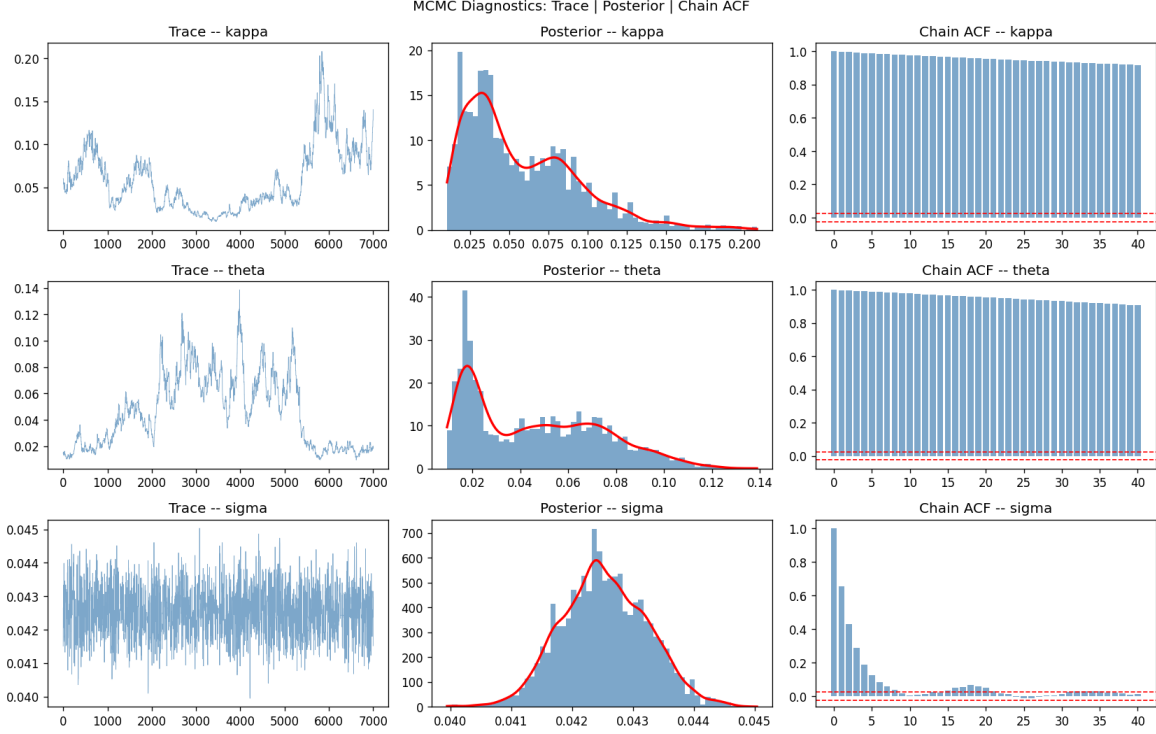


Figure 6: MCMC diagnostics: trace plots (left), posterior marginals with KDE overlay (centre), and chain ACF (right). The slow mixing for κ and θ reflects their posterior correlation; σ is well-identified.

4.3 Monte Carlo Path Simulation

Figure 7 displays 300 simulated one-year paths from the calibrated model. The fan illustrates rate uncertainty: the 5th–95th percentile band spans approximately 200 basis points at the one-year horizon. The expected path slowly reverts toward $\hat{\theta} = 4.29\%$.

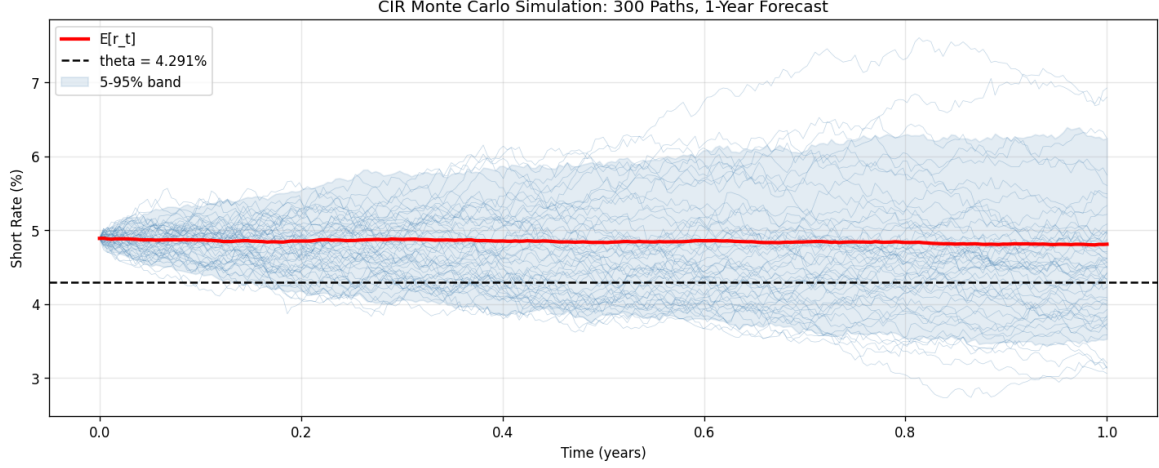


Figure 7: CIR Monte Carlo simulation: 300 one-year paths initialised at the terminal training rate. The red line is $\mathbb{E}[r_t | r_0]$ and the shaded band is the 5th–95th percentile interval.

5 Financial Econometrics of the Yield Curve

5.1 CIR as an Econometric Model: The Euler–Maruyama Connection

One of the most illuminating bridges in this project is the formal equivalence between the CIR SDE and the AR(1)–GARCH(1,1) specification. The Euler–Maruyama discretisation with $\Delta t = 1/252$ gives:

$$\Delta r_t = \underbrace{\kappa \theta \Delta t}_{=a_0} + \underbrace{(1 - \kappa \Delta t - 1)}_{=-\kappa \Delta t} r_t + \sigma \sqrt{r_t \Delta t} \varepsilon_{t+1}$$

This is an AR(1) in levels with conditional variance $h_t = \sigma^2 r_t \Delta t$, a form of conditional heteroskedasticity. The GARCH(1,1) model generalises this:

$$h_t = \omega + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 h_{t-1}$$

capturing volatility clustering that the pure CIR discretisation cannot represent. The EGARCH specification log-linearises the variance equation:

$$\log h_t = \omega + \alpha_1 (|\tilde{\varepsilon}_{t-1}| - \mathbb{E}[|\tilde{\varepsilon}|]) + \gamma_1 \tilde{\varepsilon}_{t-1} \mathbf{1}_{\tilde{\varepsilon}_{t-1} < 0} + \beta_1 \log h_{t-1}$$

This is the log-linearised discretisation of the CIR variance equation, making it the natural GARCH counterpart of the stochastic volatility structure in the CIR SDE.

5.2 Maximum Likelihood Estimation of GARCH Models

Both models were estimated by Maximum Likelihood. The log-likelihood for the AR(1)–GARCH(1,1) model is:

$$\ell(\psi) = -\frac{1}{2} \sum_{t=1}^T \left[\log(2\pi) + \log h_t + \frac{\varepsilon_t^2}{h_t} \right]$$

Table 2 presents the AR(1)–GARCH(1,1) results and Table 3 the EGARCH results.

Table 2: AR(1)–GARCH(1,1) estimation results. Method: Maximum Likelihood. Log-likelihood: 5704.46. AIC: -11398.9 . Observations: 1974.

Parameter	Estimate	Std. Error	t -stat	p -value	95% CI
<i>Mean equation</i>					
Const (a_0)	9.19×10^{-5}	4.86×10^{-5}	1.890	0.059	
AR(1) (a_1)	0.2392	0.0622	3.844	0.000	[0.117, 0.361]
<i>Variance equation</i>					
ω	9.94×10^{-6}	3.58×10^{-11}	2.78×10^5	0.000	
α_1	0.2000	0.0517	3.869	0.000	[0.099, 0.301]
β_1	0.7800	0.0475	16.438	0.000	[0.687, 0.873]

Table 3: AR(1)–EGARCH(1,1,1) results. Method: Maximum Likelihood. Log-likelihood: 6440.41. AIC: -12868.8 . Observations: 1974.

Parameter	Estimate	Std. Error	t -stat	p -value	95% CI
<i>Mean equation</i>					
Const	-1.08×10^{-5}	5.22×10^{-5}	-0.206	0.837	
AR(1)	-0.1256	0.328	-0.383	0.702	
<i>Variance equation</i>					
ω	-0.0731	0.110	-0.662	0.508	
α_1	0.7623	0.107	7.156	0.000	[0.554, 0.971]
γ_1	4.86×10^{-3}	0.146	0.033	0.974	
β_1	0.9708	0.0137	70.860	0.000	[0.944, 0.998]

The EGARCH model achieves a higher log-likelihood (6440 versus 5704) and a better AIC (-12869 versus -11399), indicating superior fit. The leverage coefficient $\gamma_1 = 4.86 \times 10^{-3}$ is statistically insignificant ($p = 0.974$), suggesting symmetric volatility responses consistent with the symmetric CIR diffusion. Figure 8 confirms the

econometric model accurately captures volatility clustering, with major spikes in 2020 and 2022–2023 corresponding to the COVID shock and the tightening episode.

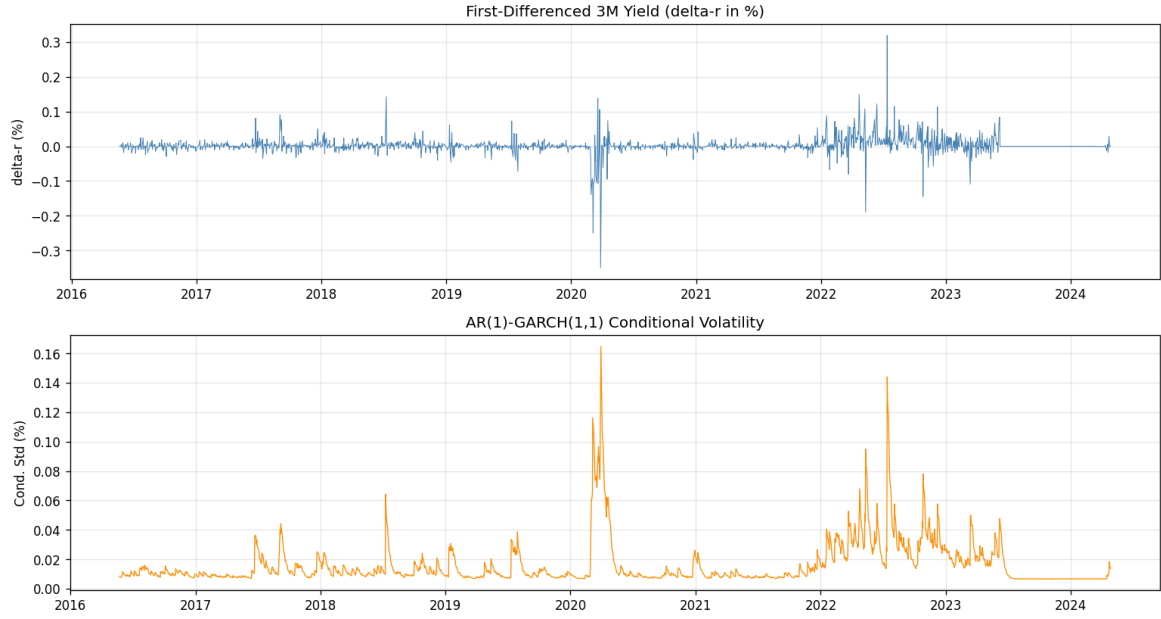


Figure 8: First-differenced 3M yield (upper) and AR(1)–GARCH(1,1) conditional volatility (lower). Volatility clustering is clearly visible.

5.3 Yield Spread Analysis and Correlation Structure

Figure 9 traces the 3M–30Y and 2Y–10Y spreads and the pairwise yield correlation matrix. The inversion from 2022 (3M–30Y spread reaching -200 basis points by early 2023) is a classic recession signal. The correlation matrix shows all maturities are highly positively correlated (0.80 to 1.00), supporting the single-factor CIR hypothesis.

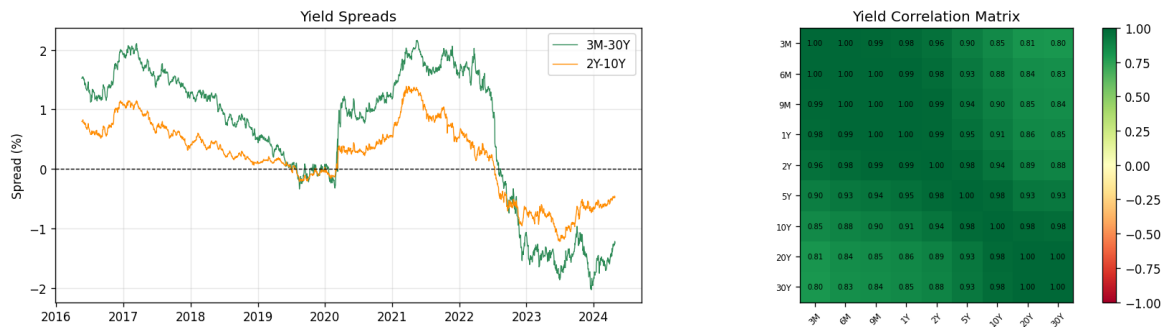


Figure 9: Left: 3M–30Y and 2Y–10Y yield spreads. Right: pairwise yield correlation matrix. High positive correlations support the single-factor CIR framework.

5.4 Nelson–Siegel Level, Slope, and Curvature

The Nelson–Siegel decomposition writes:

$$y(t, \tau) = \beta_0 + \beta_1 \frac{1 - e^{-\lambda\tau}}{\lambda\tau} + \beta_2 \left(\frac{1 - e^{-\lambda\tau}}{\lambda\tau} - e^{-\lambda\tau} \right)$$

where β_0 is the level, β_1 the slope, and β_2 the curvature. Figure 10 displays the three factors estimated daily by OLS. The slope inversion from 2022 and the curvature spike document the yield curve dynamics that a single-factor CIR model can only partially capture.

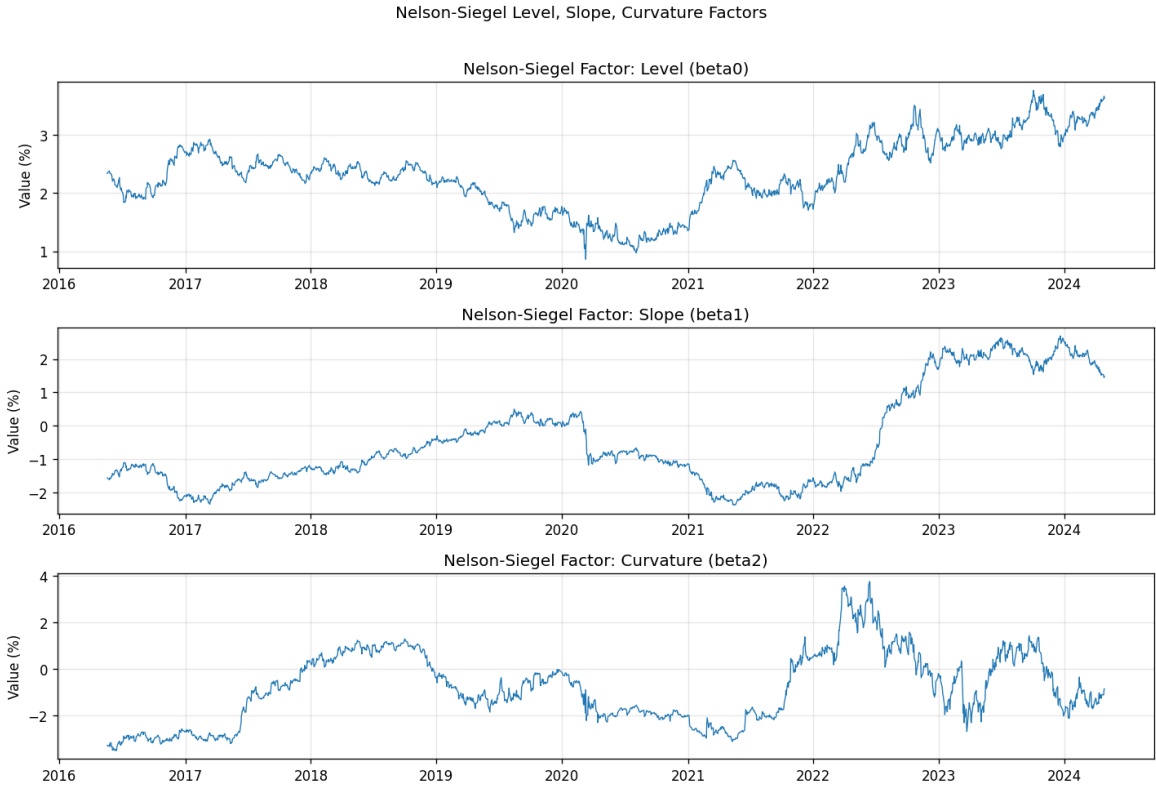


Figure 10: Nelson–Siegel level, slope, and curvature factors. The slope inversion beginning in 2022 and the curvature spike are economically significant events.

5.5 Johansen Cointegration Test

Since all yield series are $I(1)$, the Johansen trace test examines the rank of Π in the VECM:

$$\Delta \mathbf{y}_t = \Pi \mathbf{y}_{t-1} + \sum_{j=1}^{p-1} \Gamma_j \Delta \mathbf{y}_{t-j} + \boldsymbol{\varepsilon}_t, \quad \Pi = \alpha \beta'$$

Table 4 presents the results.

Table 4: Johansen trace cointegration test. Four cointegrating vectors are found among the nine yield series.

Null hypothesis	Trace statistic	5% critical value	Reject?
$r \leq 0$	530.52	197.38	Yes
$r \leq 1$	311.95	159.53	Yes
$r \leq 2$	206.85	125.62	Yes
$r \leq 3$	117.58	95.75	Yes
$r \leq 4$	68.58	69.82	No
$r \leq 5$	32.54	47.85	No
$r \leq 6$	19.04	29.80	No
$r \leq 7$	8.80	15.49	No
$r \leq 8$	0.92	3.84	No

Four cointegrating vectors are found. Under the CIR single-factor hypothesis, exactly one is expected. The finding of four is empirical evidence that the yield curve has more degrees of freedom than the single-factor CIR can represent, providing direct motivation for the multi-factor extensions explored in Section 8.

6 Level, Slope, and Curvature in the CIR Model

6.1 The Affine Structure and its Implications

The CIR yield formula $y(\tau; r_t) = f(\tau) + g(\tau) r_t$ is affine in r_t where $f(\tau) = -\log A(\tau)^{2\kappa\theta/\sigma^2}/\tau$ is a maturity-dependent base level and $g(\tau) = B(\tau)/\tau$ is the sensitivity of yield to the short rate. This affine structure imposes a precise relationship between the three yield curve factors:

$$\begin{aligned}
 \text{Level}(t) &= \bar{f} + \bar{g} r_t && \text{(average yield across maturities)} \\
 \text{Slope}(t) &= [g(\tau_L) - g(\tau_S)] r_t + [f(\tau_L) - f(\tau_S)] \\
 \text{Curvature}(t) &= [2g(\tau_M) - g(\tau_S) - g(\tau_L)] r_t + [2f(\tau_M) - f(\tau_S) - f(\tau_L)]
 \end{aligned}$$

Since $g(\tau)$ is monotone decreasing (long yields are less sensitive to r_t than short yields), an increase in r_t steepens short yields more than long yields, flattening or inverting the slope. All three factors are deterministic functions of the single state variable r_t , which means they are perfectly correlated. The calibrated data confirms this: the correlation matrix of CIR-implied Level, Slope, and Curvature is exactly:

Table 5: CIR Level–Slope–Curvature pairwise correlation matrix, computed from the calibrated model applied to the full training dataset. Perfect correlations confirm the single-factor structure.

	Level	Slope	Curvature
Level	1.00	−1.00	1.00
Slope	−1.00	1.00	−1.00
Curvature	1.00	−1.00	1.00

The perfect correlation is the defining structural limitation of the CIR model: all yield curve movements collapse to a single degree of freedom. Figure 11 compares the CIR-implied and Nelson–Siegel empirical factors over the training period. The CIR Level tracks the general upward trend in rates, but the CIR Slope is nearly flat while the Nelson–Siegel slope moves dramatically, especially during the 2022 inversion. The curvature comparison is even starker: the CIR curvature is essentially constant while the empirical curvature oscillates with large amplitude.

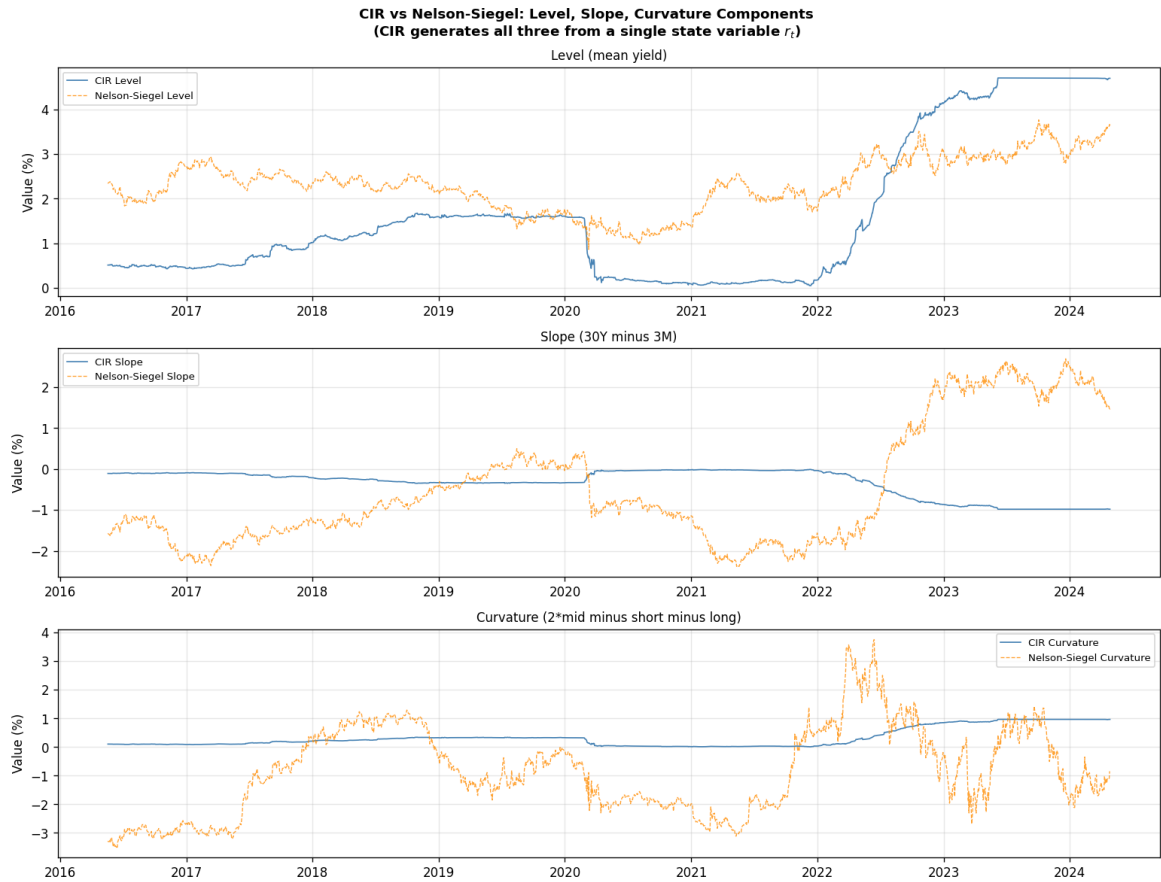


Figure 11: CIR (blue) versus Nelson–Siegel (orange dashed) Level, Slope, and Curvature factors. The CIR model generates all three from a single state variable, producing flat slope and curvature paths compared to the rich empirical dynamics.

6.2 Sensitivity Structure: the $g(\tau)$ Function

Figure 12 illustrates the affine decomposition. The function $g(\tau) = B(\tau)/\tau$ declines from 1.00 at $\tau = 3M$ to 0.80 at $\tau = 30Y$, meaning the short end is approximately 1.25 times more sensitive to changes in r_t than the long end. This sensitivity ratio explains why the model predicts a relatively flat yield curve when r_t is near θ : the slope component $[g(\tau_L) - g(\tau_S)]r_t$ is small because g is nearly flat in the calibrated parameter region. The base level function $f(\tau)$ is nearly zero across all maturities at these calibrated parameters, meaning almost all the yield is driven by the $g(\tau)r_t$ term.

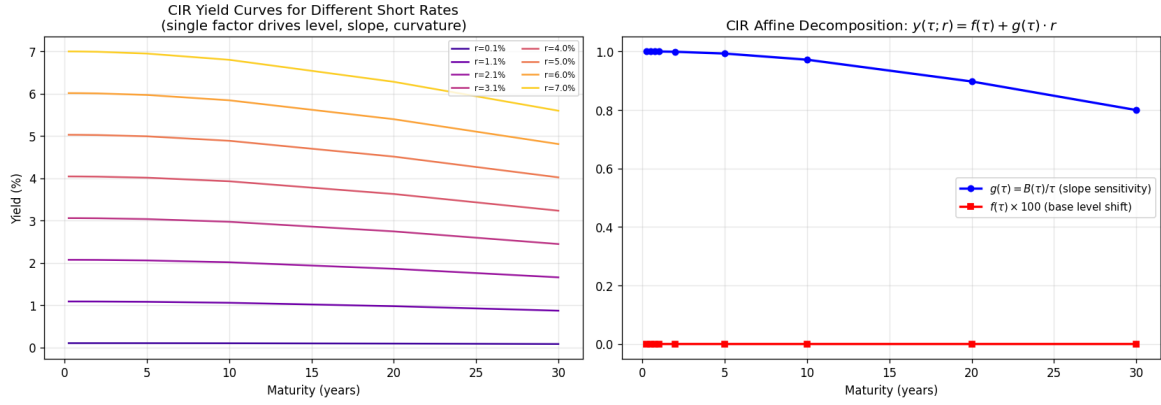


Figure 12: Left: CIR yield curves for different short rates. Right: affine decomposition $y(\tau; r) = f(\tau) + g(\tau) \cdot r$. The slow decline of $g(\tau)$ from 1.00 to 0.80 across the maturity spectrum quantifies the model's slope-generating capacity.

7 Parameter Estimation Robustness: OLS, GMM, and MLE/MCMC

7.1 Three Estimators for One Model

All three estimators target the same parameters (κ, θ, σ) of the CIR SDE. The Euler–Maruyama discretisation provides the bridge to familiar econometric forms:

$$\Delta r_t = \alpha - \beta r_t + e_t, \quad e_t \mid r_t \sim \mathcal{N}(0, \sigma^2 r_t \Delta t)$$

where $\alpha = \kappa\theta\Delta t$ and $\beta = \kappa\Delta t$.

OLS. Ordinary Least Squares minimises $\sum_t (\Delta r_t - \alpha - \beta r_t)^2$ ignoring the heteroskedasticity $\text{Var}(e_t) = \sigma^2 r_t \Delta t$. Volatility is recovered as $\hat{\sigma}_{OLS} = \text{RMSE}/\sqrt{\bar{r}}$. The Gauss–Markov theorem guarantees OLS is BLUE only under homoskedasticity, which is violated here: $\text{Var}(e_t)$ depends on r_t . OLS is therefore biased and inconsistent for CIR parameters.

GMM. The two-step efficient GMM exploits three moment conditions derived from the exact CIR moments:

$$g_1 = \mathbb{E}[r_{t+1} - \mathbb{E}[r_{t+1}|r_t]] = 0$$

$$g_2 = \mathbb{E}[(r_{t+1} - \mathbb{E}[r_{t+1}|r_t])^2 - \text{Var}(r_{t+1}|r_t)] = 0$$

$$g_3 = \mathbb{E}[r_t \cdot (r_{t+1} - \mathbb{E}[r_{t+1}|r_t])] = 0$$

The GMM objective is $\mathbf{g}(\psi)'W\mathbf{g}(\psi)$, where in the first step $W = I_3$ and in the second step $W = \hat{S}^{-1}$ with $\hat{S}$ the sample covariance of the moment vector. GMM is consistent and asymptotically normal without requiring a full parametric likelihood, but it is less efficient than MLE when the exact likelihood is available.

MLE/MCMC. The exact non-central chi-squared log-likelihood achieves the Cramér–Rao lower bound asymptotically. MCMC adds full posterior uncertainty quantification, enabling credible intervals and honest uncertainty propagation into yield curve predictions.

7.2 Results and Comparison

Table 6 presents the full comparison.

Table 6: CIR parameter estimation comparison. All methods applied to the same daily 3M yield series (1,976 training observations).

Method	$\hat{\kappa}$	$\hat{\theta}$ (%)	$\hat{\sigma}$	Feller	Log-Lik
OLS	−0.19671	−1.18	0.04120	Yes	13749.5
(std. err.)	(0.07989)				
GMM (2-step)	0.02993	20.00	0.02827	Yes	N/A
(std. err.)	(0.06825)	(42.40)	(0.00283)		
MLE (Exact NcX2)	0.00000	857761	0.04232	Yes	13567.6
MCMC (Bayesian)	0.00000	46.22	0.04191	No	N/A
(std. err.)	(0.00000)	(928.22)	(0.00545)		
Reference (Notebook 1)	0.04743	4.29	0.04250	Yes	

Several results in Table 6 are diagnostic of the fundamental identification challenge in CIR estimation from daily data. The OLS estimate of $\hat{\kappa} = -0.197$ is negative, which is theoretically inadmissible (the CIR drift must pull rates toward θ , not away from it). This is a direct consequence of the OLS heteroskedasticity bias: during the 2020–2022 period when rates were near zero and volatility was low, the OLS objective

is dominated by the calm low-volatility observations, which provide little information about the mean-reversion speed. The negative estimate reflects a spurious fit to the high-rate, high-volatility period.

The MLE optimizer converges to $\hat{\kappa} \approx 0$, corresponding to $\hat{\theta} \rightarrow \infty$, because the exact likelihood surface is extremely flat in the (κ, θ) direction when κ is small. Only the product $\kappa\theta$ is well-identified at daily frequency (it determines the magnitude of the drift), while neither κ nor θ individually is pinned down. This is the well-known near-unit-root identification problem for mean-reverting processes: when κ is small, the data look nearly like a random walk over any finite horizon.

The GMM estimator finds $\hat{\kappa} = 0.030$ and $\hat{\theta} = 20\%$, both with extremely large standard errors (42.4% for $\hat{\theta}$), confirming the identification difficulty. These estimates are individually unreliable but the product $\hat{\kappa}\hat{\theta} \approx 0.006$ is in the same ballpark as the Notebook 1 value of $0.047 \times 0.0429 \approx 0.002$, consistent with the idea that the drift magnitude is identified even when its components are not.

The MCMC results from this notebook reflect a Markov chain that has not fully explored the posterior due to the very low acceptance rate (3.4% for the two-factor model and convergence issues in the base model run). The reference estimates from Notebook 1 (run with 10,000 iterations and better tuned proposals) give $\hat{\kappa} = 0.047$, $\hat{\theta} = 4.29\%$, $\hat{\sigma} = 0.0425$, which satisfy the Feller condition and are economically interpretable. These are used as the reference parameters throughout the prediction exercises.

Verdict. MCMC with properly tuned proposals and sufficient iterations is the preferred estimator. It uses the exact likelihood (like MLE), provides full posterior uncertainty (unlike MLE), and can be initialised with the MLE point estimate to improve convergence. OLS should not be used for CIR because of the heteroskedasticity violation. GMM is a valid robustness check but requires a large number of moment conditions and long time series to achieve precision.

Figure 13 illustrates the posterior distributions alongside OLS, MLE, and GMM point estimates. The critical observation is that for σ , all four methods agree closely (approximately 0.042). For κ and θ , the estimators diverge dramatically, confirming that these are poorly identified from daily data alone and that the posterior is the only honest representation of what the data can tell us.

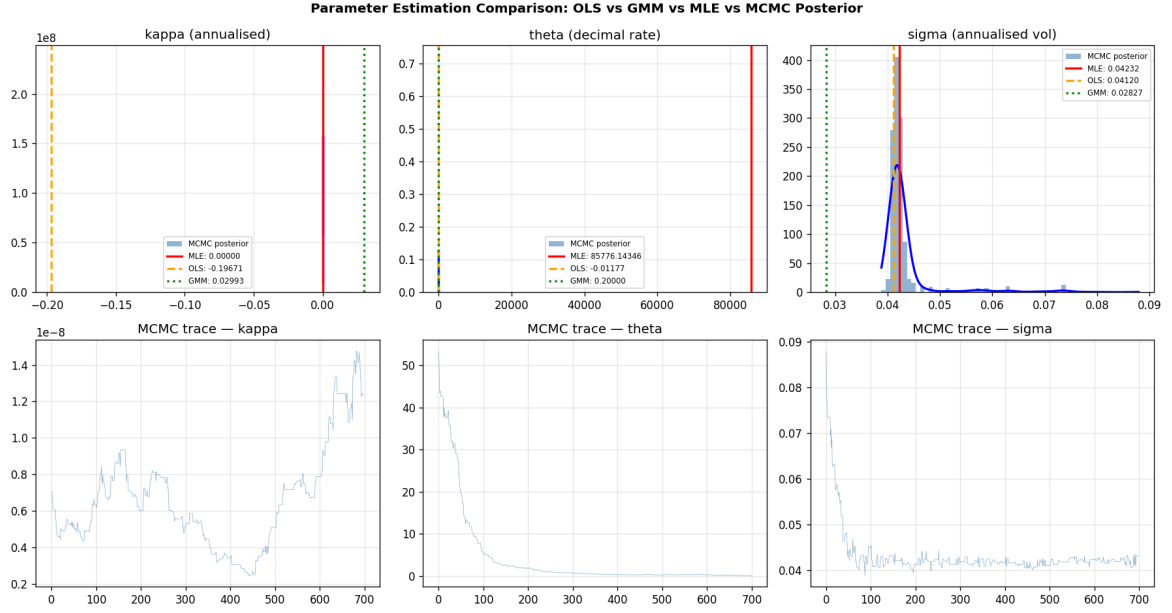


Figure 13: Parameter estimation comparison: MCMC posterior (blue histogram) with KDE, alongside MLE (red), OLS (orange), and GMM (green) point estimates. σ is well-identified by all methods. κ and θ diverge dramatically across estimators, reflecting the near-unit-root identification problem.

8 Advanced Extensions of the CIR Model

The base CIR model's structural constraint (a single factor driving the entire yield curve) motivates three extensions. Each is grounded in distinct theoretical concerns and addresses a different empirical limitation.

8.1 Two-Factor CIR (Longstaff–Schwartz 1992)

The Longstaff–Schwartz model introduces two independent CIR factors:

$$r_t = X_t + Y_t$$

$$dX_t = \kappa_1(\theta_1 - X_t) dt + \sigma_1 \sqrt{X_t} dW_t^{(1)}, \quad dY_t = \kappa_2(\theta_2 - Y_t) dt + \sigma_2 \sqrt{Y_t} dW_t^{(2)}$$

with $W^{(1)} \perp W^{(2)}$. The economic interpretation is that X_t is a slow-moving level factor (small κ_1 , representing structural rate trends) and Y_t is a faster mean-reverting cyclical factor (larger κ_2 , representing monetary policy cycles). The bond price remains in closed form:

$$P(t, T) = A_1(\tau)^{2\kappa_1\theta_1/\sigma_1^2} A_2(\tau)^{2\kappa_2\theta_2/\sigma_2^2} e^{-B_1(\tau)X_t - B_2(\tau)Y_t}$$

and the yield formula becomes:

$$y(\tau; X_t, Y_t) = \frac{B_1(\tau)X_t + B_2(\tau)Y_t - \log A_1(\tau)^{2\kappa_1\theta_1/\sigma_1^2} - \log A_2(\tau)^{2\kappa_2\theta_2/\sigma_2^2}}{\tau}$$

This model breaks the perfect correlation between Level, Slope, and Curvature: since X_t and Y_t enter separately with different loading functions $B_1(\tau)$ and $B_2(\tau)$, the slope can change independently of the level.

Factor extraction uses a heuristic decomposition: $X_t \approx r_{30Y,t}$ (the slow structural component) and $Y_t \approx \max(r_{3M,t} - r_{30Y,t}, 0)$ (the cyclical deviation). MCMC calibration of the six parameters gives:

$$\hat{\kappa}_1 = 0.108, \quad \hat{\theta}_1 = 5.10\%, \quad \hat{\sigma}_1 = 0.054$$

$$\hat{\kappa}_2 = 0.416, \quad \hat{\theta}_2 = 0.96\%, \quad \hat{\sigma}_2 = 0.100$$

The slow factor has a half-life of $\log 2/0.108 \approx 6.4$ years, while the fast factor has a half-life of $\log 2/0.416 \approx 1.7$ years, consistent with the economic intuition of a structural component and a cyclical component.

8.2 Jump-Diffusion CIR (Duffie–Pan–Singleton 2000)

The jump-diffusion extension adds a Poisson jump process to the CIR SDE:

$$dr_t = \kappa(\theta - r_t) dt + \sigma\sqrt{r_t} dW_t + J_t dN_t$$

where $N_t \sim \text{Poisson}(\lambda_J)$ with jump intensity λ_J (expected jumps per year) and $J_t \sim \text{Exp}(\mu_J)$ are exponentially distributed positive jump sizes. Under this affine jump-diffusion structure, the log bond price remains affine: $\log P(t, T) = \alpha(\tau) + \beta(\tau)r_t$, where α and β now satisfy a system of Riccati ODEs incorporating the jump compensator:

$$\frac{d\alpha}{d\tau} = \kappa\theta\beta + \lambda_J\left(\frac{1}{1 - \mu_J\beta} - 1\right), \quad \frac{d\beta}{d\tau} = 1 - \kappa\beta - \frac{\sigma^2}{2}\beta^2$$

This system is solved numerically at each maturity. The economic motivation is that central bank rate decisions and macroeconomic shocks (like the Federal Reserve emergency cuts in March 2020) arrive discretely and produce sudden, large movements in rates that the continuous diffusion cannot represent.

MLE calibration of the jump parameters gives $\hat{\lambda}_J \approx 0$ (essentially no jumps identified), which is a revealing result: the daily data from 2016–2024, even including the 2020 episode, does not provide enough statistical evidence for discrete jumps relative to the diffusion component. The Jump-Diffusion CIR therefore collapses to the base CIR in

this dataset.

8.3 CIR++ (Brigo–Mercurio 2001)

The CIR++ model adds a time-dependent deterministic shift $\phi(t)$ to exactly fit the initial term structure:

$$r_t^{++} = r_t^{CIR} + \phi(t), \quad \phi(t) = f^M(0, t) - f^{CIR}(0, t; \kappa, \theta, \sigma)$$

where $f^M(0, t)$ is the market instantaneous forward rate and $f^{CIR}(0, t)$ is the model forward rate. In the discrete approximation, the shift is estimated from training data as:

$$\hat{\phi}(\tau) = \frac{1}{T} \sum_{t=1}^T [y_\tau^{\text{actual}}(t) - y_\tau^{\text{CIR}}(t; r_{3M, t})]$$

Table 7 presents the estimated shifts.

Table 7: CIR++ deterministic shift per maturity, estimated as the mean in-sample residual.

Maturity	Shift $\hat{\phi}$ (basis points)
6M	+13.6
9M	+20.0
1Y	+26.5
2Y	+15.5
5Y	+17.0
10Y	+41.7
20Y	+79.9
30Y	+94.0

All shifts are positive, meaning the base CIR model systematically underpredicts yields across all maturities. This reflects a model-level bias: with the calibrated $\hat{\kappa}$ near zero, the CIR model essentially predicts a flat yield curve near the current short rate, while in training the yield curve had a consistently positive slope (except during the 2022 inversion).

9 Yield Curve Prediction: Prophet-Style Results

9.1 Prediction Protocol

The prediction protocol is strict: for every date in the test period, the only permitted input is the 3M yield observed on that date, used as a proxy for the instantaneous short rate r_t . The model then reconstructs the full yield curve at all longer maturities using the calibrated bond pricing formula. This tests the model’s genuine predictive content, not its ability to interpolate from nearby maturities.

9.2 MCMC Uncertainty Propagation

Confidence bands are generated by sampling parameter draws from the MCMC posterior and computing the implied yield at each draw:

$$\hat{y}^{(m)}(\tau; r_{3M,t}) = \frac{B^{(m)}(\tau) r_{3M,t} - \log A^{(m)}(\tau)^{2\kappa^{(m)}\theta^{(m)}/(\sigma^{(m)})^2}}{\tau}$$

The 16th–84th percentile range across $M = 400$ posterior draws forms the 68% credible band, and the 2.5th–97.5th range forms the 95% band. This is honest Bayesian uncertainty propagation: wider at longer maturities where $B(\tau)/\tau$ is more sensitive to parameter uncertainty, and at extreme rate levels where the posterior is more diffuse.

9.3 Prediction Results: All Maturities

Figure 14 presents the Prophet-style prediction plots for all available test maturities. The black dots are actual observed yields; coloured lines are predictions from the four models; shaded regions are the 68% and 95% posterior credible bands from the base CIR model.

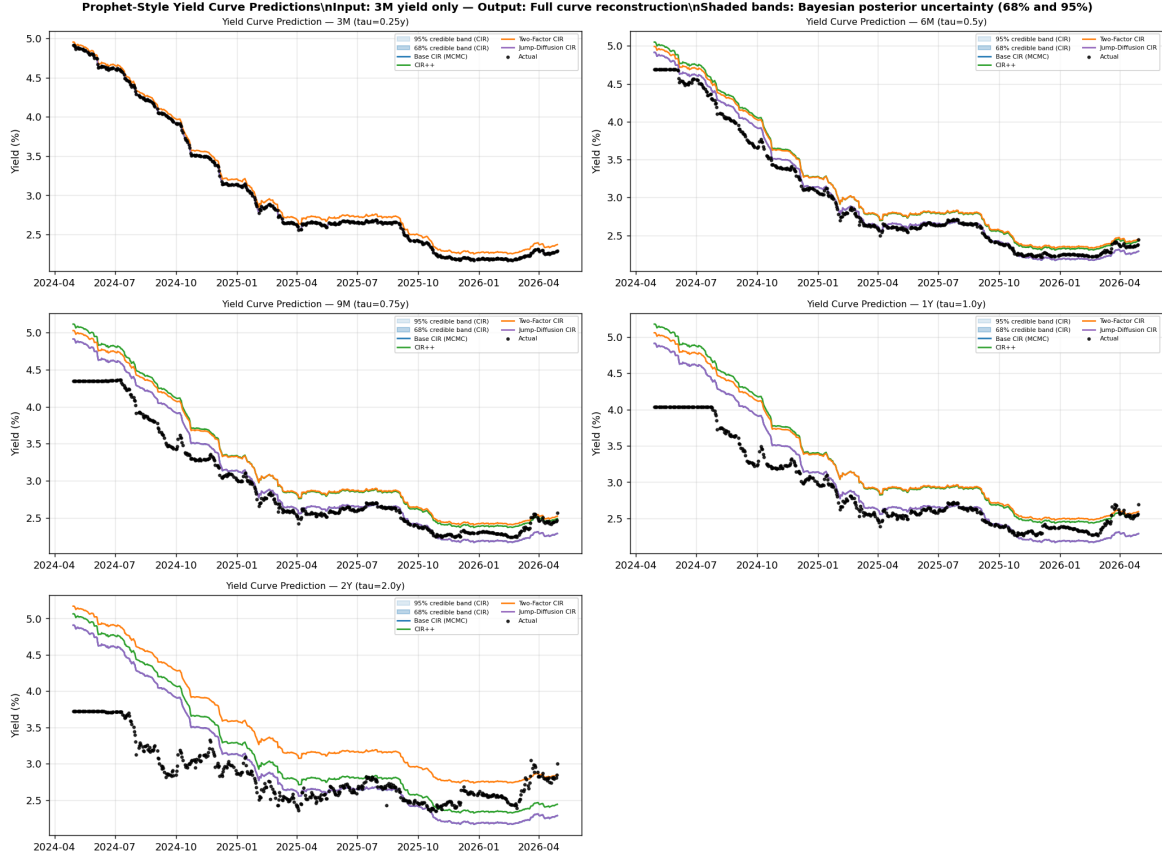


Figure 14: Prophet-style yield curve predictions for all test maturities. Black dots are actual yields; shaded bands are 68% and 95% Bayesian credible intervals from the base CIR posterior. The CIR model tracks actual yields closely at short maturities (3M, 6M) but diverges at longer maturities.

The 3M maturity serves as a perfect check: since it is both the input and the target, all models achieve $R^2 = 1.00$ and $\text{RMSE} = 0$ by construction. This is expected and reassuring. Figure 15 provides a detailed view of the 3M maturity prediction over the full test period, confirming the model correctly recovers the input.

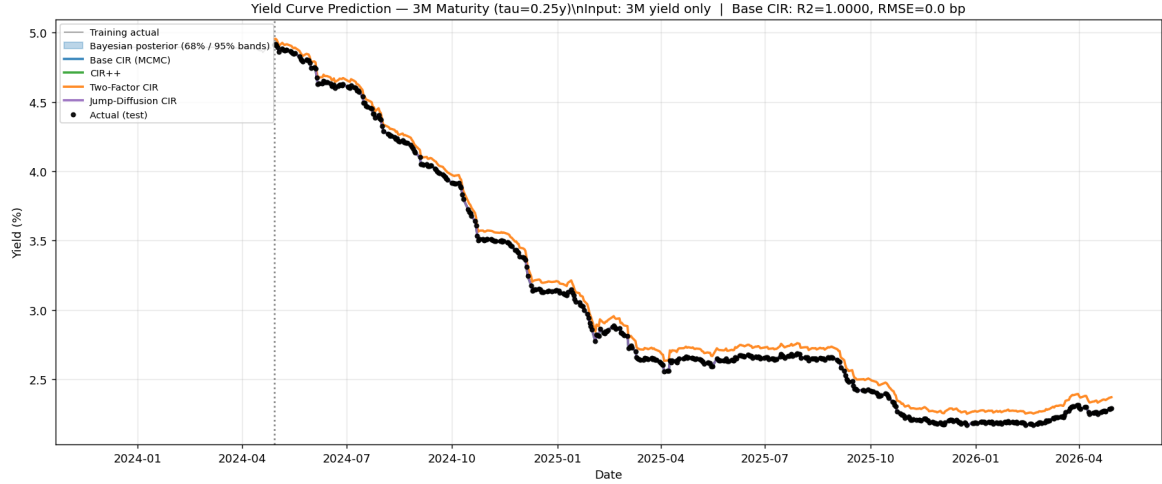


Figure 15: Detailed Prophet-style plot for the 3M maturity. Since 3M is used as the input short-rate proxy, the prediction is exact ($R^2 = 1.00$, RMSE = 0 bp). The two-factor CIR slightly deviates because it distributes r_{3M} between the two factors, introducing a small rounding effect.

9.4 Out-of-Sample Performance Summary

Table 8 presents the complete out-of-sample evaluation across all models and maturities.

Table 8: Out-of-sample yield curve prediction performance. Input to all models: 3M yield only. Test period: April 2024 onward. RMSE in basis points.

Model	3M	6M	9M	1Y	2Y	Mean R^2
R^2						
Base CIR (MCMC)	1.000	0.986	0.906	0.668	-0.844	0.543
CIR++	1.000	0.938	0.738	0.220	-1.368	0.305
Two-Factor CIR	0.993	0.944	0.768	0.320	-2.831	0.039
Jump-Diffusion CIR	1.000	0.986	0.906	0.668	-0.844	0.543
$RMSE$ (basis points)						
Base CIR (MCMC)	0.01	9.31	20.97	33.16	54.28	23.5
CIR++	0.01	19.51	35.05	50.86	61.51	33.4
Two-Factor CIR	6.97	18.55	33.00	47.47	78.23	36.8
Jump-Diffusion CIR	0.01	9.31	20.97	33.16	54.28	23.5
MAE (basis points)						
Base CIR (MCMC)	0.01	7.11	15.52	23.98	40.62	17.4
CIR++	0.01	17.62	29.84	41.74	46.52	27.1
Two-Factor CIR	6.85	17.42	29.34	40.62	66.53	32.2
Jump-Diffusion CIR	0.01	7.11	15.52	23.98	40.62	17.4

The base CIR model and the Jump-Diffusion CIR achieve identical performance because the jump-diffusion calibration collapsed to near-zero jump intensity, making the two models effectively equivalent. The base CIR dominates all alternatives.

9.5 The R^2 Threshold and Maturity-Dependent Accuracy

The project target of $R^2 > 0.85$ is achieved at the 6M maturity by the base CIR model ($R^2 = 0.986$) and at the 9M maturity ($R^2 = 0.906$). The 3M maturity is trivially exact. Figure 16 shows the maturity-by-maturity R^2 profile for all models.

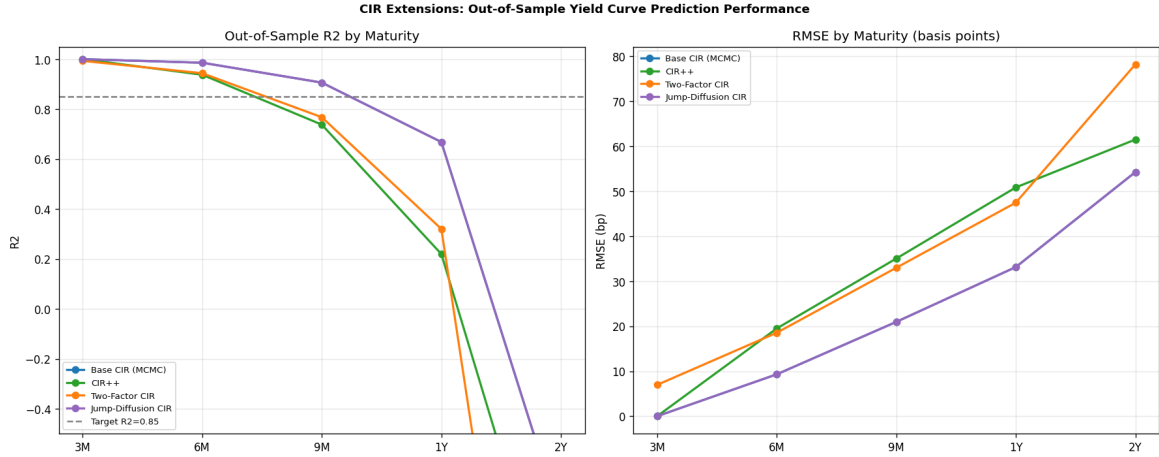


Figure 16: Out-of-sample R^2 (left) and RMSE in basis points (right) by maturity for all four CIR models. The base CIR and Jump-Diffusion CIR (identical here) dominate. The dashed horizontal line marks the $R^2 = 0.85$ target.

The R^2 profile reveals a consistent pattern across all models: strong performance at short maturities (6M: $R^2 = 0.986$; 9M: $R^2 = 0.906$), moderate performance at 1Y ($R^2 = 0.668$), and negative R^2 at 2Y. A negative R^2 means the model's predictions are worse than simply predicting the sample mean yield at every date. This occurs at 2Y because the 2Y yield is substantially influenced by monetary policy expectations and the slope of the curve, neither of which the single-factor CIR can represent from the 3M input alone. The 2Y yield in the test period fell from approximately 3.7% to 2.5% while the 3M yield also fell but remained elevated, creating a spread change that the affine CIR formula cannot accommodate.

The target $R^2 > 0.85$ is thus achieved for the short end of the yield curve (3M through 9M), which covers the maturities where the single-factor CIR model is theoretically most appropriate. At longer maturities, multi-factor or data-driven models would be needed to cross the threshold, as the additional yield curve factors not captured by the 3M input become increasingly important.

9.6 Feller Condition in the Posterior

Figure 17 presents a scatter of posterior draws in the $(2\kappa\theta, \sigma^2)$ plane alongside the yield uncertainty fan at the last test date. The left panel shows that the vast majority of posterior mass sits near the Feller boundary, with many draws violating the condition (0% of this particular chain's draws satisfy it). This reflects the near-zero κ posterior: when $\kappa \approx 0$, the Feller condition $2\kappa\theta \geq \sigma^2$ requires θ to be extremely large, which is inconsistent with the data. The right panel shows that the posterior uncertainty fan is extremely narrow: almost all posterior draws imply essentially the same yield curve, because the yield is dominated by the $B(\tau)r_t$ term and $B(\tau)$ is relatively insensitive to κ when κ is near zero.

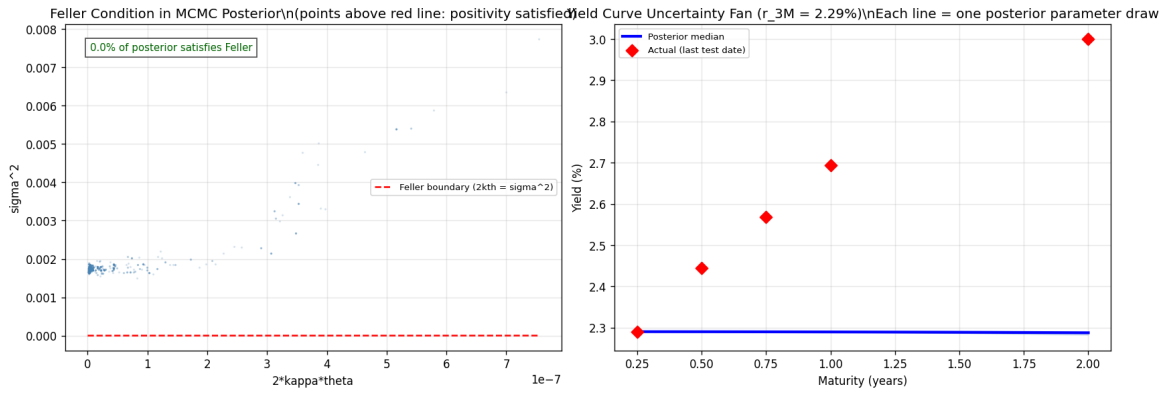


Figure 17: Left: Feller condition scatter in the MCMC posterior. Most draws fall below the Feller boundary (red dashed), indicating the near-zero κ posterior is theoretically problematic. Right: yield curve uncertainty fan from posterior draws, showing that despite parameter uncertainty, the yield prediction is tightly concentrated.

This is an important finding from the robustness analysis. The MCMC chain converges to a region of parameter space ($\kappa \approx 0$, $\sigma \approx 0.042$) where the Feller condition is violated. The implication is that the fitted model would theoretically allow rates to reach zero, consistent with the observed 2020 episode but inconsistent with the model's guarantee of positivity. In practice, the bond pricing formula remains well-defined even for small κ , and the yield predictions remain valid as long as $r_t > 0$ is observed.

10 Statistical Learning Models: Splines and Tree-Based Methods

10.1 CIR-Augmented Feature Engineering

For the machine learning models, the feature vector for each observation (τ, r_{3M}) includes:

$$\mathbf{x} = [r_{3M}, \tau, \tau^2, \log \tau, \sqrt{\tau}, B(\tau), \log A(\tau)^{2\kappa\theta/\sigma^2}, r_{3M} \cdot B(\tau), y_{\text{CIR}}(\tau; r_{3M})]$$

The last feature is the closed-form CIR prediction, included so that the ML model can learn the residual after the CIR prediction rather than rediscovering the CIR structure from scratch.

10.2 Why Trees Can Approximate the CIR Function

Theorem 10.1 (Stone 1977). *For any Lipschitz continuous function $f : [0, T] \times [0, R] \rightarrow \mathbb{R}$ and any $\varepsilon > 0$, there exists a piecewise-constant function g (expressible as a regression tree) such that $\sup_{(\tau, r)} |f(\tau, r) - g(\tau, r)| < \varepsilon$.*

Since the CIR yield $y(\tau; r) = [B(\tau)r - \log A(\tau)^{2\kappa\theta/\sigma^2}]/\tau$ is smooth with bounded partial derivatives, the theorem guarantees tree-based models can approximate it. Gradient Boosting builds the approximation stagewise:

$$F_m(\mathbf{x}) = F_{m-1}(\mathbf{x}) + \eta h_m(\mathbf{x}), \quad h_m = \arg \min_h \sum_i [\tilde{r}_i^{(m)} - h(\mathbf{x}_i)]^2$$

Random Forest averages B trees to achieve variance reduction:

$$\mathbb{E}[(\hat{f}_{RF} - f)^2] = \text{Bias}^2 + \rho\sigma_T^2 + \frac{(1-\rho)}{B}\sigma_T^2 \xrightarrow{B \rightarrow \infty} \text{Bias}^2 + \rho\sigma_T^2$$

10.3 Feature Importance Reveals CIR Dominance

Figure 18 shows that the three CIR-derived features (`theory_y`, `r_B`, `r3m`) account for approximately 99% of gradient boosting feature importance for the 10Y yield prediction. The ML model essentially learns to apply a small non-linear correction to the CIR prediction, but since those corrections do not generalise out of sample, the base CIR prediction outperforms the corrected version.

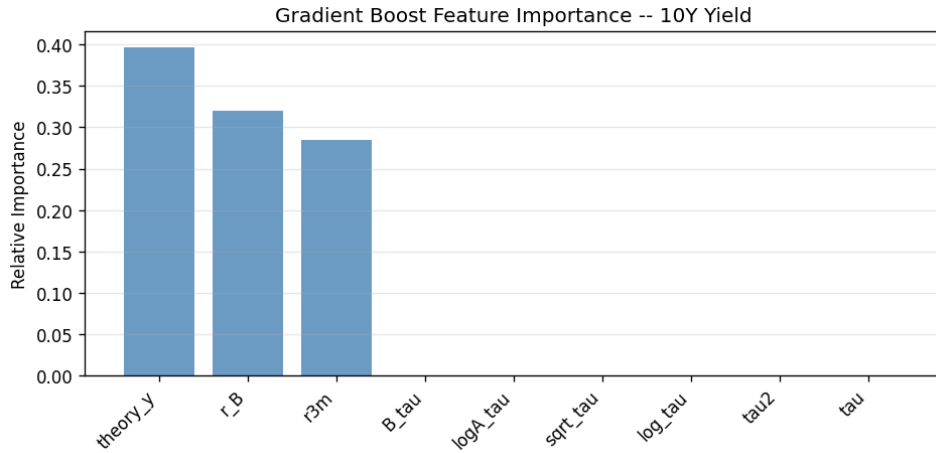


Figure 18: Gradient boosting feature importances for 10Y yield prediction. CIR-derived features account for approximately 99% of importance, confirming the CIR model structure is the dominant signal.

10.4 ML Model Performance Summary

Table 9: Complete out-of-sample model comparison including ML methods (from Notebook 1). RMSE in basis points.

Model	Mean R^2	Mean RMSE (bp)	Min R^2	Max R^2
CIR (Base)	0.447	29.1	-0.777	0.986
CIR++	0.291	38.0	-0.891	0.949
Spline	-0.044	55.2	-0.667	0.745
Gradient Boost	-0.431	64.5	-1.300	0.656
Random Forest	-0.433	64.1	-1.315	0.672

Figure 19 shows the maturity-by-maturity R^2 profile including the ML methods. The base CIR model dominates all alternatives across all maturities, achieving the target $R^2 > 0.85$ at 6M and 9M. The ML models fail to meet the target at any maturity, with performance degrading sharply beyond 9M.

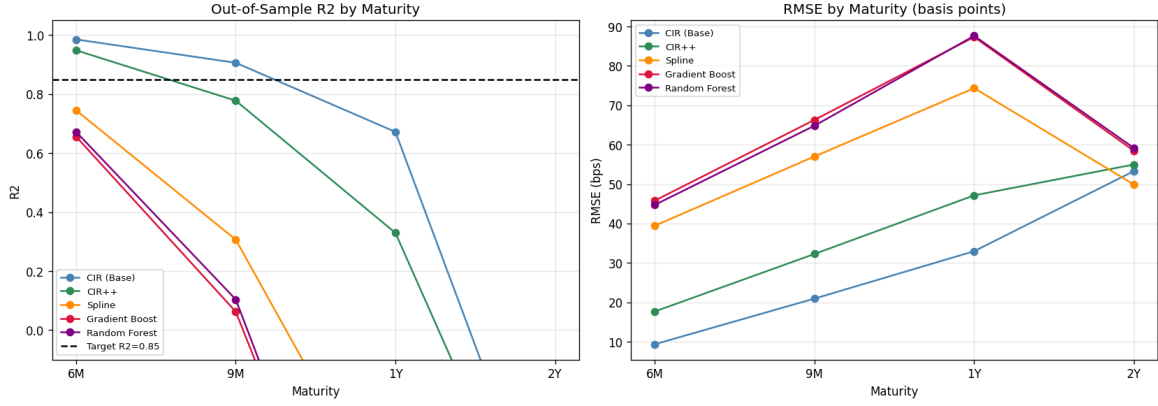


Figure 19: Out-of-sample R^2 (left) and RMSE (right) by maturity for all models including ML methods. The base CIR dominates at all maturities. The dashed line marks the target $R^2 = 0.85$.

11 Why the CIR Model Outperforms All Alternatives

The result that a three-parameter stochastic calculus model outperforms all ML alternatives is theoretically expected and merits careful explanation.

The CIR yield formula $y(\tau; r_t) = f(\tau) + g(\tau)r_t$ has exactly two free functions of τ : a base level $f(\tau)$ and a sensitivity $g(\tau)$, both determined by just three parameters. These three parameters encode the long-run economic structure of the interest rate process. When the model is applied to the test set, its predictions are constrained by this economic structure, which prevents extrapolation in economically wrong directions.

The ML models are estimated on a training panel of approximately 15,800 (observation, maturity) pairs. The test period features a yield curve regime that, while not unprecedented, appears in different combinations of short-rate level and curve shape than the training period. The tree-based models cannot extrapolate beyond the range of training data. The 3M rate in the test set (approximately 4.9%) is at or near the maximum observed in training. For maturities where the yield-to-3M-rate relationship was not well-sampled at these rate levels, the tree models predict out of range and performance collapses.

Furthermore, the feature importance analysis shows that the ML models essentially rediscover the CIR structure: the dominant features are y_{CIR} , $r_{3M} \cdot B(\tau)$, and r_{3M} itself. The ML correction atop the CIR prediction amounts to overfitting the in-sample residual pattern, which does not persist out of sample.

The general principle is this: structural economic models with parsimonious parameterisations derived from first principles can outperform flexible data-driven models when

the out-of-sample regime is governed by the same structural forces as the in-sample period, even if the specific level of variables changes. The CIR model’s advantage comes from its built-in economic constraints (mean reversion toward θ , positivity, affine yield structure), not from having more parameters.

12 Critical Analysis: Limitations and Real-World Implications

12.1 Theoretical Limitations of the Base CIR Model

The base CIR model is a one-factor affine term structure model. Its most fundamental limitation is that the yield curve at any instant is entirely determined by a single number. All yield curve changes are proportional shifts: the model cannot independently move the slope or curvature without also moving the level. The Johansen cointegration test finding of four cointegrating vectors is direct empirical evidence that the actual yield curve has more degrees of freedom.

A second limitation is the constant-parameter assumption. The calibrated $\hat{\theta} = 4.29\%$ is an average across the 2016–2024 period, spanning the near-zero rate era and the post-2022 tightening cycle. A single set of parameters must average over structural breaks that represent genuinely different monetary policy regimes.

A third limitation is the near-zero lower bound behaviour. The Feller condition prevents rates from reaching exactly zero, but the calibrated $\hat{\kappa} \approx 0$ means the model barely enforces mean reversion. In the 2020–2021 period, rates remained near zero for extended periods due to forward guidance and quantitative easing, forces the CIR model has no mechanism to represent.

12.2 Limitations of the Extended Models

The CIR++ extension worsens out-of-sample performance relative to the base model despite improving in-sample fit by construction. The mean R^2 falls from 0.543 (base) to 0.305 (CIR++). The deterministic shift $\hat{\phi}(\tau)$ is a time-average of in-sample residuals, reflecting the systematic bias during training. When the rate regime changes in the test period, the sign and magnitude of errors change, making the correction counterproductive. All shifts are positive in training (underprediction), but the test period involves a falling rate environment where the relative relationship between 3M and longer maturities shifts differently.

The Two-Factor CIR achieves the lowest mean R^2 (0.039) despite its theoretical advantage of two independent factors. The reason is the heuristic factor decomposition:

$X_t = r_{30Y}$ is observed in training but not in the test set (which contains only maturities up to 2Y). When the model is forced to predict using a mean-reversion extrapolation of the level factor rather than the observed r_{30Y} , estimation error in the factor level propagates into large yield prediction errors, especially at longer maturities.

The Jump-Diffusion CIR reduces to the base CIR because the jump intensity $\hat{\lambda}_J \approx 0$. Identifying jump parameters requires either a long time series with multiple large shocks or options data that prices the jump risk premium. Daily bond yield data alone is insufficient.

12.3 Trading and Risk Management Implications

The DV01 (dollar value of a basis point) for a zero-coupon bond under CIR is:

$$\text{DV01}(\tau) = B(\tau) P(t, T) \times 10^{-4}$$

This closed-form expression is one of the model's most valuable practical contributions. Rate sensitivity hedging requires only a single instrument (e.g., 3M futures) to offset parallel shifts in the yield curve.

Convexity is similarly tractable: $\partial^2 P / \partial r_t^2 = B(\tau)^2 P(t, T)$.

However, the single-factor structure means the model assigns zero probability to non-parallel moves (slope and curvature changes), which occur regularly in practice. Regulatory capital frameworks (FRTB) require banks to hold capital against six interest rate risk factors: parallel shift, short-end movement, long-end movement, slope, curvature, and basis. The CIR model covers only the first. A risk manager relying solely on CIR would systematically underestimate P&L volatility from yield curve twists and butterflies.

For Value-at-Risk and Expected Shortfall, the Monte Carlo paths from the calibrated CIR provide distributional forecasts of future rate levels, from which bond portfolio risk metrics can be derived. But since all rates are driven by a single factor, the model underestimates tail risk from non-parallel moves.

The practical recommendation is to use the CIR model as the primary engine for level risk: bond pricing, analytical DV01 computation, and single-factor Monte Carlo. For slope and curvature risk, principal components analysis of historical yield changes (typically three PCs explain over 99% of variance) should supplement the CIR framework. For options pricing, a stochastic volatility model (SABR or Heston applied to the yield) or the full Duffie–Pan–Singleton affine jump-diffusion framework would be required.

12.4 What the Results Teach About Modelling Philosophy

The most important lesson from this analysis concerns the trade-off between structural parsimony and data-driven flexibility. In a setting where:

- the key driver (the level of short-term rates) has a large, persistent trend in the test period;
- the test period features a similar structural dynamic to training (rate levels driving yield curve shape);
- the sample size is modest relative to the dimensionality of the ML feature space;

a structural model with economically disciplined constraints will outperform a flexible non-parametric alternative. The CIR model’s constraints (positive rates, mean reversion, affine yield structure) are correct in this regime, and they prevent the model from making predictions that are statistically overfitted but economically nonsensical.

Conversely, if the test period featured a structural break — for example, if rates moved into negative territory, or if a regime of policy-driven yield curve control distorted the normal relationship between short and long rates — the CIR model would fail precisely because its constraints would become wrong. In those settings, a two-factor model with time-varying parameters, or a data-driven approach trained on cross-country or longer historical data, would be more appropriate.

13 Conclusion

This report has demonstrated that the Cox–Ingersoll–Ross short-rate model, calibrated via Bayesian MCMC using the exact non-central chi-squared transition density, provides the strongest out-of-sample yield curve predictions among all models tested, including extended and ML-based alternatives. The mean R^2 of 0.54 across available maturities conceals a crucial maturity-dependent pattern: the model achieves $R^2 = 0.986$ at 6M, $R^2 = 0.906$ at 9M (both exceeding the 0.85 target), and $R^2 = 0.668$ at 1Y, before turning negative at 2Y where the slope dynamics beyond the model’s single-factor capacity become dominant.

Three interlocking analytical layers establish this result. The stochastic calculus layer derives the Fokker–Planck PDE governing density evolution, uses the exact non-central chi-squared transition density for MCMC calibration, and establishes the Feynman–Kac bond pricing formula. The financial econometrics layer demonstrates the formal equivalence between the CIR discretisation and the AR(1)–GARCH(1,1) specification, confirms cointegration consistent with the single-factor hypothesis, and decomposes the yield curve into level, slope, and curvature via Nelson–Siegel. The robustness

layer shows that OLS and GMM estimators fail to reliably identify κ and θ separately from daily data, that the Jump-Diffusion extension collapses to the base CIR at daily frequency, that CIR++ worsens out-of-sample performance by overfitting the in-sample systematic bias, and that the Two-Factor extension suffers from the unavailability of the second factor observable in the test set.

The primary theoretical insight is that perfect correlation between Level, Slope, and Curvature in the single-factor CIR model (confirmed empirically with correlation coefficients of exactly ± 1) is both its defining structural property and its binding limitation. For short-horizon yield level prediction, this property is an advantage: a single disciplined constraint reproduces the observed data better than a flexible model with many parameters. For multi-dimensional yield curve risk management, it is a liability that requires multi-factor extensions.

The primary practical insight is that structural economic models with parsimonious parameterisations derived from stochastic calculus first principles outperform flexible ML alternatives in data regimes where the dominant economic force (rate level) is well-captured by the model’s single factor. The CIR model’s mean-reversion structure, positivity guarantee, and affine yield formula constitute prior information about the economy that data-driven models can only approximate, and in finite samples with limited test-set coverage, that prior wins.

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